10.3 Cambial variants (Anomalous Secondary Growth)

Cambial variants (previously known as **anomalous secondary growth**) is a deviation from normal secondary growth and production of secondary vascular and non-vascular tissues. A normal cambium with abnormal activity, accessory (additional) cambia or abnormally situated cambia with normal activity can produce anomalous secondary growth. The anomalies may be listed as follows,

10.3.1 Anomalous position of vascular cambium

Unusual shapes in stems are formed by the unusual position of cambium. Example: *Thinouia scandens*, *Serjania ichthyoctona* and *Bauhinia langsdorffiana.*

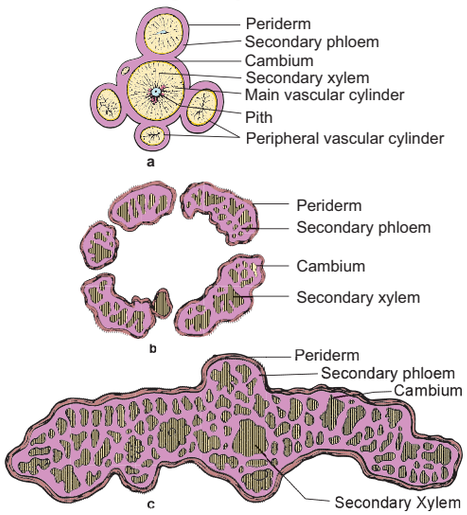


Figure 10.16: Transverse section (T.S) of stems showing anomalous scondary growth (diagrammatic) **(a)** *Thinouia scandens* **(b)** *Sarjania ichthyoctona* **(c)***Bauhinia langsdorffiana*

10.3.2 Abnormal behaviour of normal cambium

Certain segments of cambia cease to produce secondary xylem. But at the same time, it produces secondary phloem only on the outer side. The remaining cambial segments do the normal activity; As a result, ridged and furrowed steles are formed. Example: *Bignonia.*

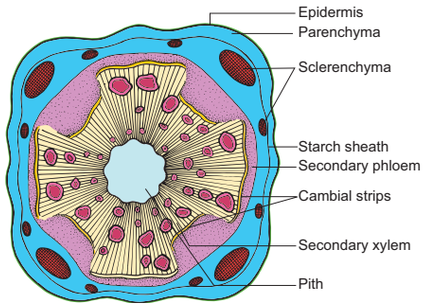


Figure 10.17: Transverse section of *Bignonia* stem (diagrammatic) showing anomalous secondary growth

10.3.3 Successive cambium

In plants like *Aviccenia, Cycas, Gnetum* a number of cambial rings arise in succession and produce concentric rings of secondary tissues.

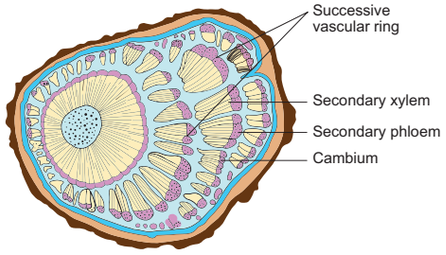


Figure 10.18: Transverse section (T.S) of *Gnetum* stem (diagrammatic) showing anomalous secondary growth

10.3.4 Interxylary or Included Phloem

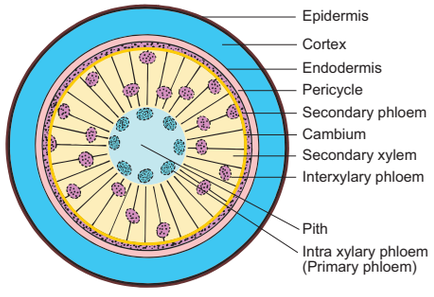


Figure 10.19: Transverse section (T.S) of *Strychnos* stem (diagrammatic) showing anomalous secondary growth

The secondary phloem found in the form of strands (islands) embedded in the secondary xylem is called **interxylary** or **included phloem**.

Example: *Strychnos, Combretum* and *Salvadora.*

10.3.5 Presence of medullary bundles along with normal cambial activity

In this type, the normal vascular bundles are arranged in a ring from which a normal cambial

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